Exact Frobenius–Hénon equalizers and their boundary defects

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Abstract

For a polynomial Hénon automorphism $H(x,y) = (y, f(y) - ax)$ over $\mathbb{F}_q$, with $\deg f = d \geq 2$ and $a \neq 0$, we compute the geometric equalizer of H^n and the coordinatewise q^r-power morphism on $\mathbb{A}^2(\overline{\mathbb{F}}_q)$. Writing $D = d^n$ and $Q = q^r$, we prove that every pair of positive clocks with $D \neq Q$ gives a finite reduced scheme with exactly $\max(DQ, Q^2)$ points. The result is uniform in every coefficient of f and in a, and holds for all pairs of clocks whenever d is not a power of the characteristic. The proof intersects projective graph closures: their total intersection number is $1+DQ+Q^2$, and the two boundary points have lengths $Q\min(D,Q)$ and 1. These exact lengths determine the short-Frobenius defect and the sharp onset of compactly supported trace agreement. They also separate two generating functions. At fixed Hénon time the Frobenius slice is a polynomial exponential times $(1-q^2t)^{-1}$, and is transcendental when $d^n > q$; a genuine diagonal iterate has a rational dynamical zeta function. A nonlinear cubic example in characteristic three has six resonant points instead of nine, showing why the equal-degree case cannot be included without further hypotheses.

Keywords: Hénon automorphism; finite field; Frobenius; intersection multiplicity; trace formula; dynamical zeta function.

1 Introduction and main theorem

Frobenius-twisted fixed points on a nonproper variety need not agree with the compactly supported cohomological trace at small Frobenius powers. For a nonlinear polynomial automorphism of the affine plane, the discrepancy can be measured by intersecting its graph with the Frobenius graph at infinity. We carry out that measurement for every iterate of a generalized Hénon map and every nonresonant Frobenius power.

Fix a prime power $q = p^e$ and put $k = \overline{\mathbb{F}}_q$. Throughout the paper,

$$H : \mathbb{A}^2_k \longrightarrow \mathbb{A}^2_k, \qquad H(x,y) = (y, f(y) - ax), \qquad f \in \mathbb{F}_q[y], \quad \deg f = d \geq 2, \quad a \in \mathbb{F}_q^*. \tag{1.1}$$

The inverse is the polynomial map

$$H^{-1}(x,y) = ((f(x) - y)/a, x). \tag{1.2}$$

There is no monicity assumption and no restriction on a lower coefficient of f. For a positive integer r, let

$$\Phi_r(x,y) = (x^{q^r}, y^{q^r}). \tag{1.3}$$

This is the actual power morphism over k, obtained by base change from the q^r-Frobenius morphism over $\mathbb{F}_q$. We use the explicit convention (1.3) throughout, independently of terminological conventions for arithmetic and geometric Galois Frobenius.

For positive integers n and r, define

$$\begin{aligned} E_H(n,r) &= \{P \in \mathbb{A}^2_k : H^n(P) = \Phi_r(P)\} \quad \text{as an equalizer scheme,} \\ M_H(n,r) &= \#E_H(n,r)(k). \end{aligned} \tag{1.4}$$

The two integers have different roles: n is the Hénon iteration clock and r is the Frobenius clock. In particular, (1.4) is not the count $\#\operatorname{Fix}(H^n)(\mathbb{F}_{q^r})$ of rational points on one fixed-period scheme. Since H is invertible, it is also the geometric fixed-point scheme of $H^{-n} \circ \Phi_r$.

Theorem 1.1 (Exact nonresonant equalizer count)**.** *Let H satisfy* (1.1)*. For every $n, r \geq 1$, the affine equalizer $E_H(n,r)$ is finite and reduced. Set $D = d^n$ and $Q = q^r$. If $D \neq Q$, then*

$$\boxed{M_H(n,r) = \max\{DQ, Q^2\}.} \tag{1.5}$$

Under the same nonresonance condition, the closures of the graphs of H^n and Φ_r in $\mathbb{P}^2_k \times \mathbb{P}^2_k$ meet properly. Their only intersections outside $\mathbb{A}^2_k \times \mathbb{A}^2_k$ are (I_-, I_-) and (I_+, I_+), where $I_- = [0:1:0]$ and $I_+ = [1:0:0]$. Their respective local intersection multiplicities are $Q\min(D,Q)$ and 1.

Corollary 1.2 (All pairs of clocks)**.** *If d is not a power of p, then* (1.5) *holds for every $n, r \geq 1$ and every map* (1.1) *of degree d over $\mathbb{F}_q$.*

The theorem supplies two connected conclusions. First, it determines the complete coefficient-uniform boundary contribution, including the regime $D > Q$ in which the affine count exceeds Q^2. Second, it turns that contribution into an exact threshold and distinguishes fixed-time Frobenius slices from the iterates of one fixed map. The equal-degree case $D = Q$ is excluded from the counting formula itself, not just from the proof's presentation.

The total projective intersection is $1 + DQ + Q^2$. The substantive local calculation is therefore the length $Q\min(D,Q)$ at the infinity point where H^n is regular. At the opposite infinity point the inverse graph is regular, and the Frobenius differential forces length one. This asymmetry explains why using only the largest affine degrees would miss the correct count.

Section 2 states the classical ownership of the trace principles used for interpretation. Sections 3 and 4 prove the theorem and corollary. Section 5 derives the exact threshold, the fixed-time slice formula and a finite-dimensional trace obstruction. Section 6 treats genuine diagonal dynamics and the resonant counterexample. All proofs of the stated results are in the main text.

2 Classical trace results and the question addressed here

Eventual trace agreement. The general sufficiently-high-Frobenius trace principle is classical. Fujiwara's work [2] is the original source used in the accounts of Shuddhodan and Varshavsky cited here. Varshavsky proves a generalization using algebraic geometry and contracting correspondences [6]. In the author manuscript arXiv:math/0505564v2, §2.2 explains the contraction obtained after a large Frobenius twist, and Theorem 2.3.2 gives the corresponding trace statement. These results place a nonproper fixed-point problem within a general theory of local terms. We do not obtain a new version of that theory. The exact count in Theorem 1.1 is instead proved directly, including powers below the eventual-agreement threshold.

Twisted étaleness and threshold growth. Shuddhodan [4] makes several distinctions relevant to the present question. In the inspected author version arXiv:1803.06461v2, Lemma 2.6 establishes twisted étaleness, Proposition 2.10 states eventual trace agreement, and Definition 2.12 with Lemma 2.14 concerns a rational cohomologically defined zeta function. Example 3.6 uses a rank-two torus to exhibit the importance of nonproperness. The paragraph immediately following that example discusses the growth of trace-agreement thresholds under iteration. Thus neither the existence of a short-twist discrepancy nor the motivation for an iteration-dependent threshold is asserted as new here. The Hénon calculation supplies the exact threshold and every defect below it, uniformly in the coefficients.

Intersection theory and cohomology. We use classical Chow intersection on a product of projective planes. The local interpretation by the length of a proper intersection of Cohen–Macaulay subvarieties is recorded in the Stacks Project, Lemma 43.16.1, tag 0B01 [5]; tag 0FEZ gives the local-complete-intersection framework. Every graph coefficient and every local length specific to H is computed below. For the later trace comparison, the cohomology of affine space and Poincaré duality are classical inputs, as in Milne [3, Example 16.3, §22 and Theorem 24.1]. Neither is an input to the geometric count.

The scope of the contribution. The question answered here concerns the full family of nonlinear maps (1.1), with both clocks left free. The local formulas at I_- and I_+ determine the equalizer even when $q^r < d^n$. The cited general trace statements and torus example do not themselves provide those two Hénon local calculations. Our contribution is this specific exact law and its consequences. No global priority claim is needed for the argument, and the bounded source check accompanying the manuscript is not an exhaustive literature review.

3 Iterate geometry and projective graph classes

Fix $n, r \geq 1$ and write

$$g = H^n = (g_1, g_2), \qquad D = d^n, \qquad Q = q^r. \tag{3.1}$$

Unless stated otherwise, all varieties, graphs and local rings are over the algebraically closed field k. Use homogeneous coordinates $[x : y : z]$ on $\mathbb{P}^2$, and let $L_\infty = \{z = 0\}$.

Lemma 3.1 (Degrees and the two infinity points)**.** *The coordinate polynomials of g satisfy*

$$\deg g_1 = d^{n-1}, \qquad \deg g_2 = D,$$

and the degree-D homogeneous part of g_2 is $L_n y^D$ for some $L_n \in \mathbb{F}_q^*$. The degree of g^{-1} is also D; its first coordinate has leading homogeneous part $K_n x^D$, where $K_n \neq 0$, and its second coordinate has degree d^{n-1}.

The projective rational extension of g is regular outside $I_+ = [1 : 0 : 0]$. It sends $L_\infty \setminus \{I_+\}$ to $I_- = [0 : 1 : 0]$ and fixes I_-. The inverse extension is regular outside I_-, sends $L_\infty \setminus \{I_-\}$ to I_+, and fixes I_+.

Proof. Let $c_d \in \mathbb{F}_q^*$ be the leading coefficient of f. For $n = 1$, $g_1 = y$ and $g_2 = f(y) - ax$ give the asserted degrees and $L_1 = c_d$. Suppose the forward assertions hold for n. Composition on the left by H gives

$$H^{n+1} = (g_2, f(g_2) - ag_1).$$

The degrees of $f(g_2)$ and ag_1 are d^{n+1} and d^{n-1}, respectively. They are unequal, so the leading term cannot cancel. The leading homogeneous part of the second coordinate is $c_d L_n^d y^{d^{n+1}}$, and $c_d L_n^d \neq 0$. This proves the forward induction in every characteristic.

For the inverse, the case $n = 1$ follows from (1.2), with $K_1 = c_d/a$. If $H^{-n} = (u_1, u_2)$ has first degree d^n, leading term $K_n x^{d^n}$, and second degree d^{n-1}, then

$$H^{-(n+1)} = ((f(u_1) - u_2)/a, u_1).$$

Its first degree is d^{n+1}, with leading term $(c_d/a)K_n^d x^{d^{n+1}}$, and its second degree is d^n. This establishes the inverse assertions without a separability condition on either coordinate polynomial.

Let G_1, G_2 be the homogenizations of g_1, g_2 to the common degree D. The projective extension is

$$[x : y : z] \longmapsto [G_1(x, y, z) : G_2(x, y, z) : z^D]. \tag{3.2}$$

Because $\deg g_1 < D$, one has $G_1(x, y, 0) = 0$, whereas $G_2(x, y, 0) = L_n y^D$. There is no indeterminacy on $z \neq 0$, since (1.1) is a polynomial morphism. On $z = 0$ the three coordinates in (3.2) vanish simultaneously exactly when $y = 0$. This is the single point I_+. For $y \neq 0$ their value is $[0 : 1 : 0]$, which also shows regularity and fixedness at I_-. Homogenizing the inverse coordinates gives, at $z = 0$, the triple $[K_n x^D : 0 : 0]$. It has base point I_- and the asserted value I_+ everywhere else on L_∞. □

Let $X = \mathbb{P}^2 \times \mathbb{P}^2$. The notation Γ_g denotes the closure in X of the graph of g on $\mathbb{A}^2$. The map Φ_r extends to the morphism

$$[x : y : z] \longmapsto [x^Q : y^Q : z^Q]$$

on all of $\mathbb{P}^2$, so its graph Γ_{Φ_r} is already closed.

Lemma 3.2 (Graph charts)**.** *The surface Γ_g is the transpose of $\Gamma_{g^{-1}}$. Over an open subset of the first factor where g is regular it is the ordinary graph of g. Over an open subset of the second factor where g^{-1} is regular it is parameterized by $Y \mapsto (g^{-1}(Y), Y)$. In particular, Γ_g is smooth at (I_-, I_-) and (I_+, I_+).*

Proof. On $\mathbb{A}^2 \times \mathbb{A}^2$, the graph of an automorphism is the transpose of the graph of its inverse. Taking closures gives the first assertion. If g is regular on an open set $U \subset \mathbb{P}^2$, its graph over U is closed in $U \times \mathbb{P}^2$, since the target is separated. The graph over $U \cap \mathbb{A}^2$ is dense in that graph, because projection identifies it with the dense open subset $U \cap \mathbb{A}^2$ of U. Thus restriction of Γ_g to the first-factor open set U is exactly that graph. Applying this reasoning to g^{-1} and transposing proves the second chart assertion. Lemma 3.1 supplies the required regular neighborhoods of I_- and I_+. Each graph chart is isomorphic to an open subset of the smooth surface $\mathbb{P}^2$. □

Write h_1 and h_2 for the pullbacks of the hyperplane class from the two factors of X. We use the classical ring and degree convention

$$A^*(X) = \mathbb{Z}[h_1, h_2]/(h_1^3, h_2^3), \qquad \deg(h_1^2 h_2^2) = 1. \tag{3.3}$$

The three codimension-two coefficients of a surface in this ring are determined by intersecting it with two generic lines on one factor or one generic line on each factor. These are intersection degrees, so inseparable fibers are counted with their scheme lengths.

Lemma 3.3 (Graph classes)**.** *In the convention* (3.3)*,*

$$[\Gamma_g] = h_1^2 + D h_1 h_2 + h_2^2, \qquad [\Gamma_{\Phi_r}] = Q^2 h_1^2 + Q h_1 h_2 + h_2^2. \tag{3.4}$$

Consequently their Chow intersection has degree

$$\deg([\Gamma_g] \cdot [\Gamma_{\Phi_r}]) = 1 + DQ + Q^2. \tag{3.5}$$

Proof. Both projections from Γ_g to $\mathbb{P}^2$ are birational, as is seen on the dense affine graph of the automorphism. Thus $\deg([\Gamma_g]h_1^2) = 1$ and $\deg([\Gamma_g]h_2^2) = 1$. These give the coefficients of h_2^2 and h_1^2 in (3.4).

For the mixed coefficient choose a line L in the first factor avoiding I_+, and a generic line L' in the second factor avoiding I_-. On L the rational extension of g is a morphism. The equation of $g^{-1}(L')$ is a generic linear combination of G_1, G_2, z^D, of degree D. Its restriction to $L \simeq \mathbb{P}^1$ is a nonzero section of $\mathcal{O}_L(D)$, and hence has zero scheme of length D. It is nonzero at the unique point $L \cap L_\infty$, because that point maps to $I_- \notin L'$. The graph chart over L is therefore sufficient to compute the mixed intersection, which is D.

For Γ_{Φ_r} the first projection is an isomorphism, giving the coefficient 1 of h_2^2. The pullback of a line by Φ_r has degree Q, giving the coefficient Q of h_1h_2. The second projection is the finite morphism Φ_r, of degree Q^2. For example, its function-field extension on the affine chart is

$$k(x^Q, y^Q) \subset k(x, y),$$

and the Q^2 monomials x^iy^j, $0 \le i, j < Q$, form a basis of the larger field over the smaller one. This yields the coefficient Q^2 of h_1^2. Here Q^2 is the degree of a finite morphism, not the cardinality of a geometric fiber: the Frobenius morphism is radicial and has one point in each geometric fiber.

Multiplying the two classes, the only surviving degree-four monomial is $h_1^2h_2^2$. Its coefficient is $1 + DQ + Q^2$, proving (3.5). We have computed a Chow product; the next section checks properness before using it as a sum of local intersection numbers. □

4 Affine reduction and the two boundary lengths

Lemma 4.1 (Reduced affine intersection)**.** *For every $n, r \ge 1$, including $d^n = q^r$, the affine equalizer $E_H(n, r)$ is finite and reduced. Every affine intersection point of Γ_g and Γ_{Φ_r} has multiplicity one.*

Proof. The equalizer is cut out in $\mathbb{A}^2$ by

$$g_1(x, y) - x^Q = 0, \qquad g_2(x, y) - y^Q = 0. \tag{4.1}$$

Since Q is a positive power of p, the two power terms have zero differential. The Jacobian of (4.1) is therefore Dg. The Jacobian matrix of H is

$$DH(x, y) = \begin{pmatrix} 0 & 1 \\ -a & f'(y) \end{pmatrix},$$

whose determinant is a. The chain rule gives $\det Dg = a^n \neq 0$ at every point. The Jacobian criterion makes the equalizer smooth of dimension zero at every solution. It is affine and of finite type over k. Its coordinate ring is consequently a zero-dimensional reduced finite-type k-algebra, hence an Artinian algebra and a finite product of copies of k. This proves finiteness and reducedness, with the empty scheme allowed at this stage. The local ring at each solution has length one. Restricting the equations of one graph to the other gives precisely (4.1), so the same length is the affine local intersection multiplicity. □

Lemma 4.2 (Location of the boundary)**.** *The set-theoretic boundary of $\Gamma_g \cap \Gamma_{\Phi_r}$ consists exactly of (I_-, I_-) and (I_+, I_+).*

Proof. Every point of Γ_{Φ_r} is $(P, \Phi_r(P))$. Its two coordinates are either both affine or both on L_∞. Suppose $P \in L_\infty$. If $P \neq I_+$, Lemmas 3.1 and 3.2 force its second coordinate on Γ_g to be I_-. In the formula $[x : y : 0] \mapsto [x^Q : y^Q : 0]$, the only point above I_- is $P = I_-$. If $P = I_+$, then $\Phi_r(P) = I_+$. There are therefore at most the two asserted boundary intersections.

Both occur. The regular fixed point $g(I_-) = I_-$ gives the first. The regular fixed point $g^{-1}(I_+) = I_+$ and the inverse-graph chart give the second. They also belong to the Frobenius graph, since both points have coordinates in $\mathbb{F}_q$. $\square$

Lemma 4.3 (Length at the regular infinity point). *If $D \neq Q$, the local intersection multiplicity at (I_-, I_-) is $Q \min(D, Q)$.*

Proof. Use the chart $y \neq 0$ in each factor, with local coordinates $u = x/y$ and $v = z/y$ centered at I_-. From (3.2), the local expression for g is

$$(u, v) \longmapsto \left(\frac{A(u, v)}{C(u, v)}, \frac{v^D}{C(u, v)} \right), \quad A(u, v) = G_1(u, 1, v), \quad C(u, v) = G_2(u, 1, v). \tag{4.2}$$

The degree calculation gives

$$A(u, 0) = 0, \qquad C(u, 0) = L_n \neq 0. \tag{4.3}$$

Thus C is a unit in the completed local ring $S = k[[u, v]]$. The Frobenius expression on this chart is $(u, v) \mapsto (u^Q, v^Q)$. Since Γ_g is the smooth graph parameterized by (u, v) here, the completed local intersection ring is

$$S/(B, v^D - v^Q C), \qquad B = A - u^Q C. \tag{4.4}$$

Multiplication by the denominator C has not changed either defining ideal, because C is invertible.

Put $m = \min(D, Q)$. If $D > Q$, then

$$v^D - v^Q C = v^Q (v^{D-Q} - C),$$

and the second factor has nonzero constant term $-L_n$. If $D < Q$, then

$$v^D - v^Q C = v^D (1 - v^{Q-D} C),$$

and the second factor has constant term 1. Hence in both nonresonant cases the ideal in (4.4) is exactly (B, v^m). By (4.3),

$$B(u, 0) = -L_n u^Q. \tag{4.5}$$

Let $R = S/(B)$. We next verify that v is a non-zero-divisor in R. Suppose $vh = Bb$ in S. Reducing modulo v gives $0 = -L_n u^Q b(u, 0)$ in the domain $k[[u]]$. Thus $b(u, 0) = 0$, so $b = vb_1$ for some $b_1 \in S$. Cancelling v in the domain S gives $h = Bb_1$. This proves the claimed injectivity in R.

For each $i \geq 0$, multiplication by v^i now induces an isomorphism

$$R/vR \longrightarrow v^i R / v^{i+1} R. \tag{4.6}$$

Surjectivity follows from the definition of $v^i R$. For injectivity, if $v^i h = v^{i+1} b$ in R, injectivity of multiplication by v^i gives $h = vb$. Equation (4.5) identifies

$$R/vR \simeq k[[u]]/(u^Q),$$

which has basis $1, u, \ldots, u^{Q-1}$ and length Q. The finite filtration of $R/v^m R$ by the images of $R, vR, \ldots, v^{m-1}R$ has m successive quotients of this length. It follows that

$$\operatorname{length}_k S/(B, v^m) = mQ = Q \min(D, Q). \tag{4.7}$$

In particular the local intersection is isolated. Completion preserves the length of a finite-length local module, so (4.7) is also the length of the uncompleted local intersection ring. $\square$

Lemma 4.4 (Length at the inverse-regular infinity point)**.** *For every $n, r \geq 1$, the local intersection at (I_+, I_+) is transverse and has length one.*

Proof. Parameterize Γ_g near (I_+, I_+) by the second coordinate, as $(g^{-1}(Y), Y)$, using Lemma 3.2. In this chart the equation for intersection with Γ_{Φ_r} is

$$Y = \Phi_r(g^{-1}(Y)). \tag{4.8}$$

Choose affine coordinates s, t on $x \neq 0$ centered at I_+. The inverse is regular and fixes the origin in these coordinates. The morphism Φ_r is $(s, t) \mapsto (s^Q, t^Q)$ and has zero differential. Therefore the two entries of the left side of (4.8) minus the right side have linear terms s and t, respectively.

In $k[[s, t]]$, let J be the ideal generated by those two entries and let $\mathfrak{m} = (s, t)$. Their linear terms imply $\mathfrak{m} = J + \mathfrak{m}^2$. Applying Nakayama's lemma to the finite module $\mathfrak{m}/J$ gives $\mathfrak{m}/J = 0$. Hence $J = \mathfrak{m}$ and $k[[s, t]]/J = k$, of length one. The independent linear terms also prove transversality. □

Proof of Theorem 1.1. Lemma 4.1 proves affine finiteness and reducedness for all positive clocks. Assume now $D \neq Q$. Lemma 4.2 exhausts the boundary support, and Lemmas 4.3 and 4.4 show that both boundary intersections are isolated. The two projective surfaces in the smooth fourfold X therefore meet properly, in dimension zero.

The Frobenius graph is isomorphic to $\mathbb{P}^2$ and is smooth everywhere. The graph Γ_g is smooth at every intersection point: on the affine graph this follows from the first projection, and at the two boundary points it follows from Lemma 3.2. The local rings of both surfaces are consequently Cohen–Macaulay. The classical proper-intersection length formula applies at each point [5, Lemma 43.16.1, tag 0B01]. Thus the lengths just computed are the local multiplicities in the Chow product.

Subtracting the two boundary contributions from (3.5), and using that each affine point has length one, gives

$$\begin{aligned} M_H(n, r) &= 1 + DQ + Q^2 - Q\min(D, Q) - 1 \\ &= DQ + Q^2 - Q\min(D, Q) \\ &= \max(DQ, Q^2). \end{aligned}$$

The same calculation establishes the boundary assertions in the theorem. □

Proof of Corollary 1.2. If $d^n = q^r = p^{er}$ for positive integers n, r, every prime divisor of d must be p. Hence d is a power of p. Under the corollary's hypothesis this is impossible, so every pair of clocks satisfies the nonresonance condition in Theorem 1.1. □

5 The exact trace threshold and fixed-time slices

For this section assume that d is not a power of p. This hypothesis ensures that the count is available at every positive Frobenius power for any fixed n, with no missing resonant coefficient in a generating series.

5.1 Compactly supported trace and its exact defect

Fix a prime $\ell \neq p$. The isomorphism H^{-n} and the finite morphism Φ_r are proper, so their composite acts by pullback on compactly supported étale cohomology. Define

$$T_H(n, r) = \sum_{i=0}^{4} (-1)^i \operatorname{Tr}\Big((H^{-n} \circ \Phi_r)^*; H^i_c(\mathbb{A}^2_k, \mathbb{Q}_\ell)\Big). \tag{5.1}$$

The classical cohomology computation gives

$$H_c^i(\mathbb{A}_k^2, \mathbb{Q}_\ell) = \begin{cases} \mathbb{Q}_\ell(-2), & i = 4, \\ 0, & i \neq 4. \end{cases} \tag{5.2}$$

Indeed, acyclicity of affine space and Poincaré duality give (5.2); see the inputs cited in Section 2. A polynomial automorphism has degree one and acts as one on this top group. The power morphism has degree Q^2 and acts as Q^2. This can also be computed on $\mathbb{P}^2$: Frobenius pulls a hyperplane class back to Q times itself, and hence its square to Q^2 times itself; the top compactly supported group of $\mathbb{A}^2$ identifies with that top projective group. Thus

$$T_H(n, r) = q^{2r}. \tag{5.3}$$

This cohomological calculation interprets, but does not prove, the geometric count.

Proposition 5.1 (Sharp trace-agreement threshold)**.** *For each fixed $n \geq 1$, put*

$$R_n = \left\lfloor \log_q(d^n) \right\rfloor. \tag{5.4}$$

Then

$$M_H(n, r) - T_H(n, r) = \begin{cases} d^n q^r - q^{2r} > 0, & 1 \leq r \leq R_n, \\ 0, & r \geq R_n + 1. \end{cases} \tag{5.5}$$

The first positive integer at which equality occurs, and the first integer from which equality always holds, are both $R_n + 1$. In particular, this threshold equals $n \log_q d + O(1)$ as $n \to \infty$.

Proof. The number $\log_q(d^n)$ is not an integer, since an integer value would give a resonance excluded by Corollary 1.2. Thus $q^r < d^n$ holds exactly for positive integers $r \leq R_n$, and $q^r > d^n$ holds exactly for $r \geq R_n + 1$. Combining Theorem 1.1 with (5.3) gives (5.5). The inequality is strict at every integer before $R_n + 1$, and equality holds at every later integer, proving both threshold statements. Finally, $R_n + 1 = n \log_q d + \theta_n$ with $0 < \theta_n < 1$, which proves the stated asymptotic without an unspecified coefficient-dependent error. □

For example, the all-clock degree family contains every quadratic map (1.1) in odd characteristic and every cubic map of that form in characteristic two. Proposition 5.1 computes an exact source-system threshold in these families. It does not assert that iteration-dependent thresholds or eventual trace agreement are new phenomena.

5.2 Fixed-Hénon-time Frobenius slices

Definition 5.2. For fixed $n \geq 1$, the Frobenius slice generating function is

$$\mathcal{Z}_{H,n}(t) = \exp\left(\sum_{r \geq 1} M_H(n, r) \frac{t^r}{r} \right) \in \mathbb{Q}[[t]]. \tag{5.6}$$

The word "slice" records that n stays fixed while r varies.

Proposition 5.3 (Slice formula and transcendence)**.** *For every $n \geq 1$,*

$$\mathcal{Z}_{H,n}(t) = \frac{\exp(P_n(t))}{1 - q^2 t}, \qquad P_n(t) = \sum_{r=1}^{R_n} (d^n q^r - q^{2r}) \frac{t^r}{r}. \tag{5.7}$$

If $d^n < q$, then $P_n = 0$ and the slice is rational. If $d^n > q$, then the slice is transcendental over $\mathbb{C}(t)$.

Proof. By (5.5), the logarithm in (5.6) is the sum of

$$\sum_{r\geq 1} q^{2r}\frac{t^r}{r} = -\log(1-q^2t)$$

and the finite polynomial $P_n(t)$. Exponentiating the formal identity gives (5.7). If $d^n < q$, then $R_n = 0$ and the finite sum is empty.

Suppose $d^n > q$. Then $R_n \geq 1$, and every coefficient appearing in P_n is positive by (5.5). In particular P_n is a nonconstant real polynomial with positive leading coefficient. We give an elementary proof that $\exp(P_n)$ is transcendental. If it were algebraic over $\mathbb{C}(t)$, clearing the denominators of an algebraic relation would give

$$\sum_{j=0}^{J} b_j(t)\exp(jP_n(t)) = 0, \qquad b_j \in \mathbb{C}[t], \quad b_J \not\equiv 0, \tag{5.8}$$

with $J \geq 1$. A relation holding as a formal series at zero holds analytically near zero, because each term is entire, and hence holds everywhere by the identity theorem. For all sufficiently large positive real t, divide (5.8) by $b_J(t)\exp(JP_n(t))$ to obtain

$$1 + \sum_{j=0}^{J-1}\frac{b_j(t)}{b_J(t)}\exp(-(J-j)P_n(t)) = 0.$$

Each rational quotient has at most polynomial growth on this ray. Since $P_n(t) \to +\infty$ with positive polynomial leading term, each summand after the 1 tends to zero. The limit would be $1 = 0$, a contradiction. Thus $\exp(P_n)$ is transcendental. Multiplying it by the nonzero rational function $(1-q^2t)^{-1}$ preserves transcendence, proving the assertion about the slice. □

The defining equations in (1.4) change with r. Definition 5.2 therefore does not start from the rational points of one fixed variety. The classical fixed-variety rationality theorem of Dwork [1] imposes no rationality conclusion on this construction. When $d^n < q$, the resulting function does happen to coincide with the Hasse–Weil zeta function of $\mathbb{A}^2$; when $d^n > q$, Proposition 5.3 proves its transcendence. The construction also differs from the cohomologically defined zeta function discussed in Section 2. Its defining clock must be specified even when two resulting rational functions coincide.

5.3 A finite-dimensional trace obstruction

Proposition 5.4 (No invertible-Frobenius weighted trace for a defective slice)**.** *Fix $n \geq 1$ with $d^n > q$. There do not exist finitely many finite-dimensional vector spaces V_i over a characteristic-zero field, endomorphisms A_i and invertible endomorphisms F_i of V_i, and signs $\epsilon_i \in \{1, -1\}$ such that*

$$M_H(n,r) = \sum_i \epsilon_i \operatorname{Tr}(F_i^r A_i) \quad \textit{for every } r \geq 1. \tag{5.9}$$

No commutation condition on F_i and A_i is imposed.

Proof. Suppose (5.9) holds. For each i, write the characteristic polynomial of F_i as

$$\chi_i(z) = z^{m_i} + c_{i,m_i-1}z^{m_i-1} + \cdots + c_{i,0}.$$

Invertibility of F_i implies $c_{i,0} \neq 0$; zero-dimensional spaces may be omitted. By Cayley–Hamilton, $\chi_i(F_i) = 0$. Multiplying on the left by F_i^r, on the right by A_i, and taking the trace shows that $u_{i,r} = \mathrm{Tr}(F_i^r A_i)$ satisfies the recurrence with characteristic polynomial χ_i for all $r \geq 1$. This operation does not move A_i past F_i, which is why no commutation is required.

Let $\mathcal{E}$ denote the forward shift of a sequence indexed by $r \geq 1$. The polynomial

$$\chi(z) = (z - q^2) \prod_i \chi_i(z)$$

has nonzero constant coefficient and annihilates

$$v_r = \sum_i \epsilon_i u_{i,r} - q^{2r} = M_H(n, r) - q^{2r}.$$

Indeed, each summand is annihilated by one of the factors of $\chi(\mathcal{E})$, and the polynomial shift factors commute. Writing $\chi(z) = b_0 + b_1 z + \cdots + b_L z^L$ gives

$$b_0 v_r + b_1 v_{r+1} + \cdots + b_L v_{r+L} = 0, \qquad r \geq 1, \qquad b_0 \neq 0. \tag{5.10}$$

By Proposition 5.1, $v_r = 0$ for every $r \geq R_n + 1$. Substituting $r = R_n$ into (5.10) therefore gives $v_{R_n} = 0$. Repeating with $r = R_n - 1$ and then decreasing r one at a time proves $v_r = 0$ for every $r \geq 1$. But $v_1 = d^n q - q^2 > 0$, a contradiction in a characteristic-zero field. $\square$

The obstruction is both finite-dimensional and all-clock. It permits eventual trace agreement, as (5.5) already demonstrates. It also leaves infinite-dimensional constructions and explicit boundary corrections outside its conclusion.

6 Diagonal iteration and the resonant boundary

6.1 One fixed dynamical system

Continue to assume that d is not a power of p. For positive integers b, s, define the single polynomial self-map

$$S = H^{-b} \circ \Phi_s : \mathbb{A}_k^2 \longrightarrow \mathbb{A}_k^2. \tag{6.1}$$

The exponent m in S^m now advances both original clocks, in the fixed ratio $(n, r) = (bm, sm)$. This is different from keeping n fixed.

Proposition 6.1 (Diagonal point counts and dynamical zeta)**.** *For every $m \geq 1$, the fixed-point scheme of S^m is finite and reduced, and*

$$\#\,\mathrm{Fix}(S^m)(k) = \Lambda^m, \qquad \Lambda = \max\{q^{2s}, q^s d^b\}. \tag{6.2}$$

Hence its Artin–Mazur zeta function, defined by the fixed-point counts, is

$$\zeta_S(t) = \exp\left(\sum_{m \geq 1} \#\,\mathrm{Fix}(S^m)(k) \frac{t^m}{m}\right) = \frac{1}{1 - \Lambda t}. \tag{6.3}$$

Its compactly supported cohomological determinant is instead $(1 - q^{2s} t)^{-1}$; these two rational functions agree if $d^b < q^s$ and differ if $d^b > q^s$.

Proof. All coefficients of H lie in $\mathbb{F}_q$, so for each $s \geq 1$,

$$f(y^{q^s}) = f(y)^{q^s}, \qquad a^{q^s} = a.$$

Substitution into (1.1) shows $H \circ \Phi_s = \Phi_s \circ H$. The inverse H^{-1} is also defined over $\mathbb{F}_q$, or one can compose this equality with H^{-1} to obtain commutation for the inverse. It follows by induction on m that

$$S^m = H^{-bm} \circ \Phi_{sm}.$$

The equation $S^m(P) = P$ is equivalent, as an equalizer scheme, to $H^{bm}(P) = \Phi_{sm}(P)$ after applying the automorphism H^{bm}. Theorem 1.1 and Corollary 1.2 therefore give

$$\begin{aligned}\#\,\mathrm{Fix}(S^m)(k) &= M_H(bm, sm)\\ &= \max\{d^{bm}q^{sm}, q^{2sm}\}\\ &= \Big(\max\{d^b q^s, q^{2s}\}\Big)^m.\end{aligned}$$

They also give finiteness and reducedness. Substituting this count in the formal logarithm of the zeta gives $\sum_{m\geq 1} \Lambda^m t^m/m = -\log(1-\Lambda t)$, which proves (6.3).

By (5.2), only the top compactly supported group contributes to the cohomological determinant. The pullback by S acts there as q^{2s}, since H^{-b} acts as one. Thus the alternating determinant is $(1-q^{2s}t)^{-1}$. There is no equality $d^b = q^s$ under the degree hypothesis. In the smaller-degree regime $\Lambda = q^{2s}$; in the larger-degree regime $\Lambda = d^b q^s > q^{2s}$. □

There is no conflict with an eventual trace theorem for a fixed correspondence. The powers S^m change the Hénon correspondence as well as its Frobenius twist. If $d^b > q^s$, this diagonal sequence stays on the short-twist side of the threshold at every m. The rational function (6.3) is a determinant identity for this source dynamical system, with no proposed identification with an external arithmetic Euler factor.

6.2 A nonlinear resonant counterexample

We now allow d to be a power of p, solely to test the excluded case. The nonresonance condition was used at a precise point in Lemma 4.3. If $D = Q$, the second local equation is

$$v^D - v^Q C = v^Q(1-C). \tag{6.4}$$

The factor $1-C$ can fail to be a unit. The formula for its local length is then no longer determined by the two degrees alone through the preceding argument.

Proposition 6.2 (Failure at equal degrees)**.** *Over $\mathbb{F}_3$, let*

$$H(x,y) = (y, y^3 + y^2 - x). \tag{6.5}$$

For $n = r = 1$ one has $d^n = q^r = 3$, but

$$M_H(1,1) = 6 \neq 9 = \max\{dq, q^2\}.$$

Proof. The equations $H(x,y) = (x^3, y^3)$ are

$$y = x^3, \qquad y^3 + y^2 - x = y^3.$$

After substituting the first equation, the second becomes $x^6 - x = 0$. Thus the coordinate ring of the equalizer is $k[x]/(x^6 - x)$. In characteristic three the derivative of $x^6 - x$ is -1. This degree-six polynomial has six distinct roots over k, each of which determines exactly one $y = x^3$. The count is six. □

The term y^2 in (6.5) makes this a nonadditive polynomial example. Affine reducedness is still guaranteed by Lemma 4.1; it is the degree-only formula at the boundary, not reducedness, that fails. A classification of resonant coefficients would require additional local information and is not part of Theorem 1.1.

7 Scope and reproducibility

The exact equalizer law follows from three source-specific ingredients: pure leading terms for every Hénon iterate, a constant nonzero affine Jacobian, and the two opposite regular graph charts at infinity. The local length $Q\min(D,Q)$ accounts for the whole short-twist defect. The resulting count depends only on d,n,q,r, even though the theorem allows every lower coefficient and every nonzero value of a. That coefficient independence is useful for uniformity, but limits what the observable can distinguish between maps.

Limitations. The formula is nonresonant. It does not classify the equal-degree coefficients, the ordinary periodic-point zeta function of H, or the Galois groups of exact-period schemes. The two-clock count is taken over the algebraic closure, not over one chosen finite extension. Its arithmetic is intrinsic to the source coefficients and Frobenius action. No target Euler factors, root numbers, automorphy, functional equation, target zero-divisor correspondence, or Hilbert–Pólya realization is established. The finite-dimensional obstruction in Proposition 5.4 has the stated invertibility and all-clock hypotheses and makes no infinite-dimensional assertion.

Exact checks and data availability. The accompanying package contains the unchanged program `bounded_check.py` and its saved output `bounded_results.json`. The program constructs (4.1) over prime fields and counts the standard monomials of a Gröbner basis to obtain the affine quotient dimension. It separately verifies the determinant a^n of the defining Jacobian. The recorded execution used Python 3.12.3 and SymPy 1.14.0. Its 47 nonresonant checks comprise all 36 quadratic coefficient/Jacobian choices over $\mathbb{F}_3$ at $(n,r)=(2,1)$ and eleven additional choices covering both degree regimes, higher iterates, characteristic two, and nonprime base fields with prime-subfield coefficients. The saved resonant negative control returns six instead of nine. These are finite exact checks of examples, not a proof of the quantified theorem. No additional computation was performed for this manuscript.

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